Use an oven to bake and recycle barbie dolls



Learn to Code with C++



How we found a WIIIIIIIDE Wii

Tech Monthly

Issue 2

Evolution of windows

Find out the evolution of Windows and where things started out and what went wrong for Microsoft…

Halloween Projects

Get the best out of Halloween with **Lego Mindstorms, Raspberry Pi** and **Arduino**! Make your own Halloween electronic projects to scare people off!

How Did We Get Here?



Anthony Hardes

Anthony is editor of Tech Weekly and is a tech hobbyist. His latest project is located:

Legolandgame .neocities.org

Technology evolves at a rapid pace. New innovations emerge and often, what seems to be an innovation now, becomes a relic of the past after a couple of years. Operating systems change with time, hardware advances, and the way we use our computers changes too.

In this issue, we'll have a look at new innovations, long-gone user interfaces, and current technologies and software.

Our main feature will look at the long-gone, beautiful Windows Aero graphic user interface, used in Windows Vista and Windows 7, and will investigate where it came from, what made it so special, and how its aesthetic evolved over time.

We'll also have other articles related to technology, computer gaming, and more.

Brief News

Microsoft to Drop Snapdragon 850 Support

Microsoft has confirmed that Windows 11 26H2 will be the final version of Windows to officially support Qualcomm's Snapdragon 850 processor. The chip was used in some of the earliest Windows-on-ARM laptops, but Microsoft is now moving on from the older hardware. PCs using the chip can still run Windows 11 26H2 and receive support until October 2028. After that, owners won't officially be able to upgrade to the next major version of Windows. It's another sign of how quickly Microsoft's Windows-on-ARM platform is moving forward, leaving some of its earliest hardware behind.

Microsoft may be preparing a major Windows 11 platform update for 2027

The 2026 update will allow different versions of Windows 11 to be run on the same platform, so that users can choose which version they want to use.

The change is significant because Windows 11 26H1 and 26H2 will be based on different platforms. Recent reports suggest that 27H2 may be the first major platform update for Windows 11 since 25H2 to be available to the broader Windows 11 user base. Microsoft has not yet officially released Windows 11 27H2, but early versions of next-generation Windows have already been made available to Windows Insider participants. If the reports are correct, 2027 may be a major year for Windows 11.

GNOME 50.5 Released With Bug and Security Fixes

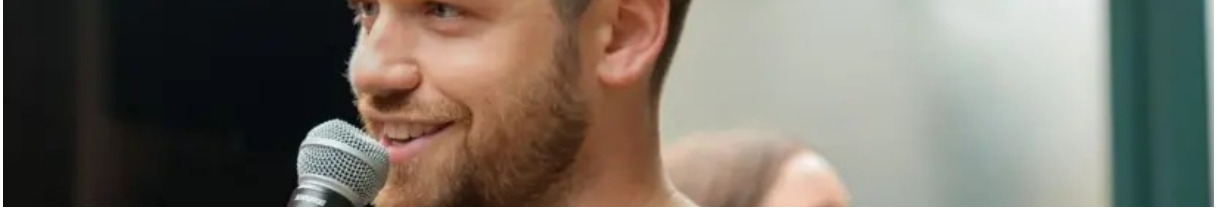
The GNOME Project has announced the release of GNOME 50.5, a bug fix and security update for the GNOME 50 desktop environment. Released on September 24, this update is primarily focused on resolving crashes, memory leaks and security problems, rather than introducing new features. The update touches upon 22 components, including GNOME Shell, Mutter, GTK, GDM, GVfs and Epiphany. Among the fixes are issues related to workspace switching, external monitors, wireless connections and lock screens. A number of security problems were addressed as well, including vulnerabilities affecting GVfs, Epiphany and librsvg. GNOME recommends that users of GNOME 50 upgrade to the latest packages from their Linux distribution.

Raspberry Pi OS Updated With Battery and Squeekboard Dock Widgets

Raspberry Pi has released a new Raspberry Pi OS update featuring multiple improvements to its desktop environment. The update introduces Battery and Squeekboard widgets to the new dock, simplifying the ability to view battery information and access the on-screen keyboard.The update also includes support for automatically mounting encrypted drives, improvements to Control Centre, higher quality Raspberry Pi menu icons and several minor fixes.The latest Raspberry Pi OS is built using Debian 13 "Trixie" and utilises the Linux 6.18 LTS kernel.

Data Centre Set to Replace Microsoft Headquarters

Plans have been submitted for a £200 million data centre on the site of Microsoft's former UK headquarters at Thames Valley Park in Earley, near Reading. Three of the former office buildings are due to be demolished to make way for the new facility. The proposal for the data centre would have around 72MW of computing capacity which could require up to 110MW of electricity. As the local electricity network currently only has 1.8MW available for the site, the developers intend to generate some of this power using natural gas fuel cells, pending the upgrades to the network. The project has been met with mixed reactions, with concerns about noise, emissions and its impact on the surrounding area. However, developers state that the site is suitable for the UK's increasing requirements for digital infrastructure, and a planning decision is set for 2027.



Design Studio turns Barbie dolls into Tech

How would you dispose of an out-of-use Barbie doll that is no longer being used? One design studio in Brighton has come up with an unusual solution: create technology. Brighton-based design studio gomi has partnered with Mattel's Barbie brand to form the Childhood Memories Collection. The plastic material of unwanted Barbie and Ken dolls can be salvaged and repurposed into new products such as Bluetooth speakers, power banks, wireless chargers and tracker tags.

The collection launched in September 2026.The idea came from gomi co-founder and chief designer Tom Meades, who graduated from the University of Brighton with a degree in 3D Design and Craft in 2017, who previously focussed on turning scrap into technology, meaning that the Barbie collaboration is a natural extension of what the company already does. From Dolls to Devices. The interesting feature of the collection is that the link to Barbie isn't just on the outside - material recovered from Barbie dolls, Ken dolls and their accessories is incorporated into the products. The recovered plastic is mixed with other reclaimed materials, including difficult-to-recycle plastic waste, and turned into colourful marbled surfaces.This gives the products a very different appearance from most modern technology, where the usual plastic is either black, white or grey, while the speakers and other devices have bright patterns that can be different from one item to another, with some designs being made as individual pieces, meaning you aren't simply buying another identical gadget.



Figure 1: Recovered materials from the dolls and accessories are incorporated into a marbled design

The Bluetooth speaker is one of the main products of the collection, which uses recycled Barbie materials for its casing and has an aluminium back plate and buttons. It can connect using Bluetooth or a 3.5mm auxiliary connection and has a claimed battery life of 32 hours. Gomi claims the speaker is also splash resistant and its design allows it to be repaired rather than simply thrown away when something breaks. The collection also features power banks, which are designed to be useful for charging phones and other devices, away from a plug socket. Gomi claims its power banks use repurposed e-bike battery cells, giving another type of discarded material a second life.

Designed to Last. One of the biggest ideas behind the collection is that recycling shouldn't be the only factor to be considered when designing a product, but that gomi also wants its technology to last for as long as possible.Many modern electronic products can be difficult to repair, with the entire product being replaced if a battery stops working or other components fail. Gomi takes a different approach, where its products are designed to be opened and repaired, with screws being used to hold components together rather than glue, allowing individual parts to be replaced instead of requiring the whole product to be thrown away. Gomi also offers repair and upgrade services, with the company stating its products are made to last, rather than being short-lived gadgets that are replaced with something newer. This is important with electronics, as the manufacturing process of new devices requires new materials and energy.

Why Barbie? At first, turning Barbie dolls into electronics might seem like a strange idea, but the collaboration works because Barbie is something that many people associate with childhood. An old doll might have spent years in a bedroom, cupboard or box before eventually being forgotten, but instead of treating the plastic as useless waste, gomi is giving it another purpose, with a piece of a childhood toy potentially ending up as part of a speaker sitting on someone's desk years later. There is also something interesting about the contrast between the original toys and their

new purpose, with Barbie being designed as a toy and a gomi speaker or power bank being designed for everyday technology. The material has gone from something used for playing to something that is actually used for modern digital life. Gomi is also encouraging people to send in old Barbie dolls that they no longer want, which can be used in future products, but the idea is not to encourage people to throw away dolls that are still being enjoyed, but to find another use for toys that have already reached the end of their useful life.

Making Technology More Interesting. Another interesting part of the project is the design, where many technology products are designed to be simple and neutral. Phones, chargers, speakers and computers often come in black, white, silver or grey. Gomi is going the opposite way, wanting technology to have more personality and the recycled Barbie plastic makes every product look slightly different. This also means that the recycling process is part of the design rather than something hidden away, as the waste can be seen in the finished product, being part of what makes the product look interesting. The Barbie collection also combines nostalgia, recycling and product design, as it takes something that could have been considered waste and turns it into something useful, keeping some of the visual character of the original material. It also shows that sustainable design doesn't need to be boring, as the recycled materials can become part of the appearance of a product, not being hidden away, with gomi's view that an old Barbie shouldn't be the end of the story, but could become a speaker, charger or other piece of technology used for years to come.



This article is inspired by BBC News' article about the same topic. This is our main source when researching this article. You can view the news article on: https://www.bbc.co.uk/news/articles/cv1j30djp1yno Alternatively scan QR code

IMAGE CREDIT: Gomi Studios

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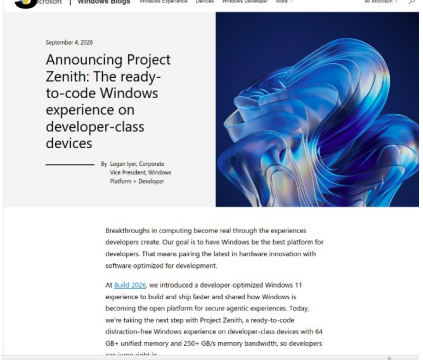
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THE MAIN HYPE



New Microsoft Project Zenith has been announced as a programmer friendly version of Windows 11. It will also provide a “distraction-free windows experience” for developers. Once it will be released, it will be available to the public and not just developers. See more.

Blogs.windows.com



September 4, 2026

Announcing Project Zenith: The ready-to-code Windows experience on developer-class devices

By Logan Iyer, Corporate Vice President, Windows Platform + Developer

Breakthroughs in computing become real through the experiences developers create. Our goal is to have Windows be the best platform for developers. That means pairing the latest in hardware innovation with software optimized for development.

At Build 2026, we introduced a developer-optimized Windows 11 experience to build and ship faster and shared how Windows is becoming the open platform for secure agentic experiences. Today, we're taking the next step with Project Zenith, a ready-to-code distraction-free Windows experience on developer-class devices with 64 GB+ unified memory and 250+ GB/s memory bandwidth, so developers

WHAT IS ON TODAY?



Raspberry Pi locks down Pi 5 RAM upgrades in firmware. Raspberry Pi added a feature in their firmware that restricts users from swapping RAM chips, sometimes even RAM chips of the same capacity from other Raspberry Pi boards. See more.

The walkthrough of how our weekly digest pulls those into one dashboard

- Pricing for your size (20 seats) and the two-week onboarding timeline

Jeffgeerling.com



Jeff Geerling

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Raspberry Pi locks down Pi 5 RAM upgrades in firmware

Sep 21, 2026

Raspberry Pi added a feature in their firmware that restricts users from swapping RAM chips, sometimes even RAM chips of the same capacity from other Raspberry Pi boards.

Apparently there were people buying cheap 1 or 2 gig Pi 5s, soldering in some cheap and sometimes unreliable 8 gig RAM chips, and selling them as new 8 gig Pi 5s.

Now I totally understand where Raspberry Pi's coming from. They don't wanna support Pis being sold like this, because flaky RAM and dodgy resellers are a huge problem. Especially with how profitable can be in the current AI bubble.

But I also totally disagree with the way they're handling the situation.

The firmware update that disabled RAM swaps was actually from late 2024, two years ago. So you can't blame the RAM pricing situation for this.

Raspberry Pi is within their legal rights to lock down their hardware like this, but that doesn't mean we have to like it.

*We don’t publish daily... Yet

The Fake GTA VI demo that was malware

By **Anthony Hardes**



Figure 1 credits: Leon Neal / Getty images

Grand Theft Auto VI is one of the most anticipated games of the year, and it seems that cybercriminals want to use this to their advantage. In August 2026, security researchers noted the appearance of websites with a false Rockstar Games theme, promising a playable GTA VI demo. However, it turned out that clicking on the “Play Now” button led to the installation of a malicious Windows application on the user’s computer, not the desired game demo. Phishing websites that looked official. Rockstar’s phishing sites were convincing, as they used real information about GTA VI’s upcoming promotional campaigns. However, it was not hard to notice that the promised demo was about 1.1 megabytes — which was suspiciously light for a modern AAA game. Malwarebytes researchers have concluded that the “demo” was actually an information stealer called Vidar, part of a malware family known as Vidar Stealers. Information stealer malware: what is it capable of? Instead of the expected game,

GTA VI phishing websites distributed malware that could steal sensitive information stored in the browsers, such as passwords, cookies, and sessions.The last of these could allow unauthorized access to a large extent, even with two-factor authentication, since sessions often contain access tokens. ([TechCrunch][1])Thus, users who wanted to try the GTA VI demo received a chance to become victims of cybercrime.### The fact that there is no GTA VI demo at allThe most obvious aspect of all this, as noted by Malwarebytes, is that Rockstar Games does not provide a playable demo of GTA VI. In fact, several security researchers pointed out that the entire event was an attempt to mislead users by holding out information about the real launch of the game. ([Malwarebytes][2])It is noteworthy that GTA VI’s hype has already been used several times by cybercriminals in the past months. Thus, in early 2026, fraudsters offered early access to GTA VI, while in mid-2026, another scam was reported, promising a pirated version of GTA VI. ([Malwarebytes][3])### The lesson: always check the sourceThe GTA VI malware incident demonstrates how dangerous an object of desire can be. With millions of people wanting to know the first details about the game, criminals only have to take advantage of the situation by distributing malicious software under the guise of a demo. Thus, if an unknown website offers an early version of the game, it is most likely a fake — and it is safest to avoid suspicious sources altogether.The most reliable way to obtain information about the release of the game is to turn to the official website of its developers, Rockstar Games. However, until its release, users are advised to be wary of scams that lure in unwary users with promises of early access to GTA VI. A fake demo could steal your session, and you might find yourself logged into someone’s social media account.Unknown sources that claim to have early access to GTA VI should be treated with caution. This could be the beginning of a dangerous path for your personal data.



This article is sourced from trusted websites. These websites are: Malwarebytes and techcrunch.com.

Part 2

iPhone 18… I have very mixed feelings

The camera isn’t everything. The camera is probably the most important feature Apple is boasting about for the upcoming iPhone 18. Year after year we’re introduced to better sensors, advanced processing and other enhancements, but at what point do we realize our photos don’t need to get any better? Don’t get me wrong, the camera quality on the iPhone 18 is fantastic and noticeably better than any previous models, but the jump in quality between year to year is beginning to shrink. If you upgrade to the newer model from a few years back, then yes, you’ll probably see a major improvement in your photographs. But if you own an iPhone 16 or 17, I’m not sure how much you’ll actually benefit from upgrading. And this is what I mean when I say that cameras have gotten so good, that we’ve hit a point where the improvements are barely noticeable. You can make the resolution higher, allow more light in, and make colours more accurate, but in the end you’re still just taking a picture.

Should you actually upgrade? This is precisely what I've been asking myself throughout this article. The iPhone 18 isn't a bad phone. In fact, there are plenty of things that I like about it. However, it is a question whether that is enough for an upgrade. If you have an old iPhone, the iPhone 18 would be a great upgrade path for you. You'll get newer processors, better battery technology, improved cameras and years of software support ahead of you. On the other hand, if you already have a relatively recent iPhone, things start to get complicated. You could be spending a lot of money for a phone that feels surprisingy similar to the one that you already have. That's what troubles me most about the state of the smartphone industry. We are reaching the point where the technology is improving faster than the reasons for a smartphone upgrade. My final th

After using the iPhone 18 and having in mind all that I've said, I have very mixed feelings about this product. There are definitely improvements, but I don't think that every improvement is something that everyone will actually notice. Battery life matters. Performance matters. A good camera matters. But so does the experience of actually using the phone. For me, the iPhone 18 doesn't feel like a phone that completely changes what an iPhone can be. It feels more like another step forward in a very long line of improvements. And that's okay. The real question isn't "Is the iPhone 18 better?" - it's "Is it better enough for you to upgrade?”



Figure 1: Apple’s “Revolutionary” camera upgrade on the iPhone 11

The things Apple doesn't talk about. But there's another side to the iPhone 18 Pro that is harder to notice from the spec sheet. Apple has made improvements to things like AI, the Action Button and Camera Control, but these aren't necessarily features that will completely change how you use your phone. The A20 Pro chip is perhaps more interesting for what it could do in the future than what it does today. Current apps and games don't necessarily need all of this power, but having extra performance available means the phone has more room to handle increasingly demanding software over the next few years. The camera is another interesting example. Apple has introduced a variable aperture, but the physical main camera sensor hasn't dramatically changed. What's more interesting is that the iPhone 18 Pro can produce more detailed images than the 17 Pro even with the variable aperture disabled. This suggests that Apple's image processing and the A20 Pro's processing power are playing a major role in the improvement. Apple has also improved Photographic Styles, allowing users more control over things such as colour, skin tone, smoothing and film grain. There are also more customisable professional camera

controls, allowing people to choose which settings they actually want to use. So perhaps the iPhone 18 Pro isn't about one enormous new feature. Instead, it's about lots of smaller improvements happening behind the scenes. And that brings me back to my original question - is it actually worth upgrading? If you're coming from an older iPhone, there are plenty of reasons to consider it. But if you're already using an iPhone 17 Pro, the improvements may be much harder to notice. The iPhone 18 Pro can be a better phone without necessarily being a better reason to upgrade. That, more than anything, is why I have such mixed feelings about it.





iPhone 18 vs iPhone 17

Specs at a Glance



Feature	iPhone 18 Pro Max (Rumored)	iPhone 18 Pro (Rumored)	iPhone 17 Pro Max	iPhone 17
Chip	A20 Pro (2nm)	A20 Pro (2nm)	A19 Pro (3nm)	A19 (3nm)
RAM	12GB	12GB	12GB	8GB
Display	6.9" LTPO+ OLED	6.3" LTPO+ OLED	6.9" LTPO OLED	6.3" LTPO OLED
Main Camera	48MP variable aperture	48MP variable aperture	48MP f/1.78 fixed	48MP f/1.6 fixed
Telephoto	48MP periscope (5x)	48MP periscope (5x)	48MP periscope (5x)	None
Front Camera	24MP (under-display Face ID)	24MP (under-display Face ID)	18MP	18MP
Battery	~5,200mAh	TBD	5,088mAh	3,692mAh
Dynamic Island	Smaller (partial under-display)	Smaller (partial under-display)	Standard	Standard
Release	September 2026	September 2026	Available now	Available now
Starting Price	~$1,199	~$1,099	$1,199	$799

Note: iPhone 18 specs are based on supply chain reports and analyst predictions. Apple has not officially confirmed any details.

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Orange Pi 4A: The Raspberry Pi Clone with Eight Cores



When you think of a small single-board computer, chances are you think of Raspberry Pi. However, Raspberry Pi isn't the only company that makes small single-board computers. The Orange Pi 4A is one of the many Raspberry Pi clones, and one of the many products in an attempt to offer something different than the Raspberry Pi. At first glance, it looks like exactly what you'd expect from a Raspberry Pi-style computer: a small circuit board with GPIO pins, USB ports, HDMI, networking and a microSD card slot. But it also has some interesting hardware of its own beneath the surface.## What's inside?Orange Pi 4A uses an Allwinner T527 processor with eight ARM Cortex-A55 cores running at up to 1.8GHz. The board has optional 2GB or 4GB of

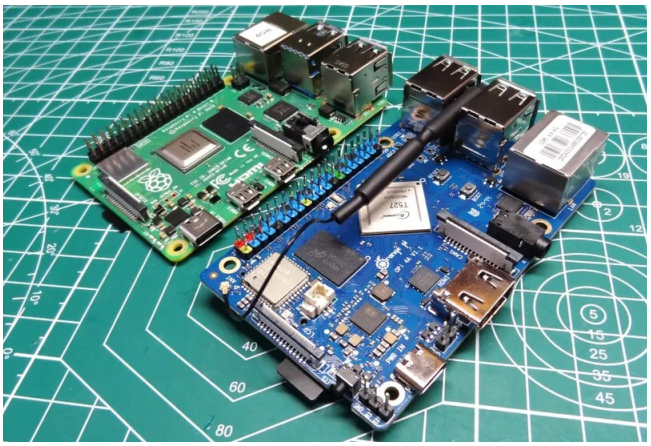


Figure 1: The Raspberry Pi 4 compared to the Orange Pi 4A

LPDDR4/4X memory, along with a Mali-G57 GPU and 2TOPS NPU, for AI-related workloads. One of the largest advantages the Orange Pi 4A has over a traditional Raspberry Pi is its storage capabilities. Alongside the microSD card slot, it also includes an M.2 slot for an NVMe SSD, and eMMC storage as an option. This includes considerably more storage options than the Raspberry Pi 4 would normally have. It also includes Gigabit Ethernet, Wi-Fi 5, Bluetooth 5.0, HDMI 2.0 and a 40-pin expansion connector. In short, it looks like a pretty serious little computer.## How does it perform?This is where the first issue appears. In testing performed by Tom's Hardware, Orange Pi 4A's multi-core performance was considerably stronger than the Raspberry Pi 4's. Its eight cores enabled it to reach a Geekbench 6 multi-core score of 1,033 compared to only 719 for the Raspberry Pi 4. However, its single-core score was considerably lower. In terms of day-to-day performance, the picture isn't nearly as impressive. The reviewer found the operating system generally fairly responsive, but at times sluggish, with applications such as GIMP taking longer to launch than on the Raspberry Pi 4. Video playback was also a weakness in their testing. YouTube playback at 1080p suffered significant dropped frames and a 4K local video test performed particularly poorly. So while eight cores may sound impressive, more cores don't necessarily guarantee a better experience everywhere.## The NVMe advantage. The M.2 NVMe slot is probably one of the Orange Pi 4A's most interesting features. Having an NVMe SSD provides much faster storage than a typical microSD card could offer, and it's supported by the board, which supports PCIe 2.0 NVMe drives. However, there's an important caveat: in their testing, Tom's Hardware found that the NVMe drive couldn't be used as the boot drive. It's still possible, but you have to mount the drive and use it after the operating system has booted up. That's a slight disappointment as being able to boot directly from NVMe would make the feature much more useful.## And then there's the heat...Orange Pi 4A also seems to run noticeably hotter than the Raspberry Pi 4.In Tom's Hardware's testing, the board idled at around 54°C, compared to 38.4°C for the Raspberry Pi 4. Under a five-minute stress test, the Orange Pi 4A reached around 75°C.

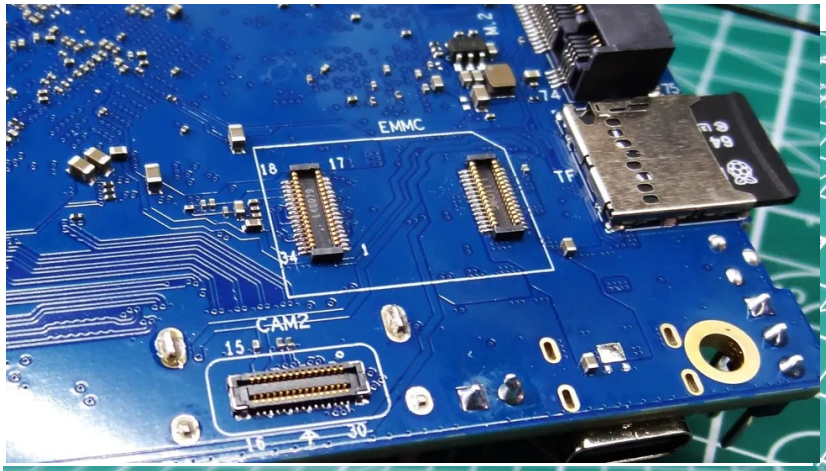
Orange Pi 4A: The Raspberry PI clone with eight cores!

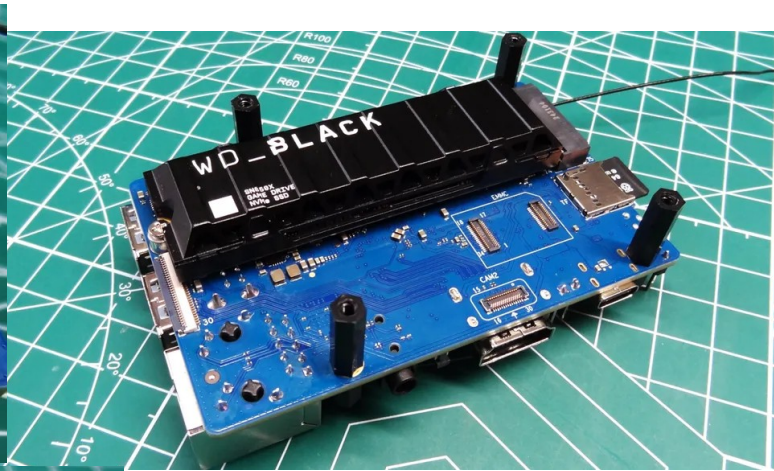
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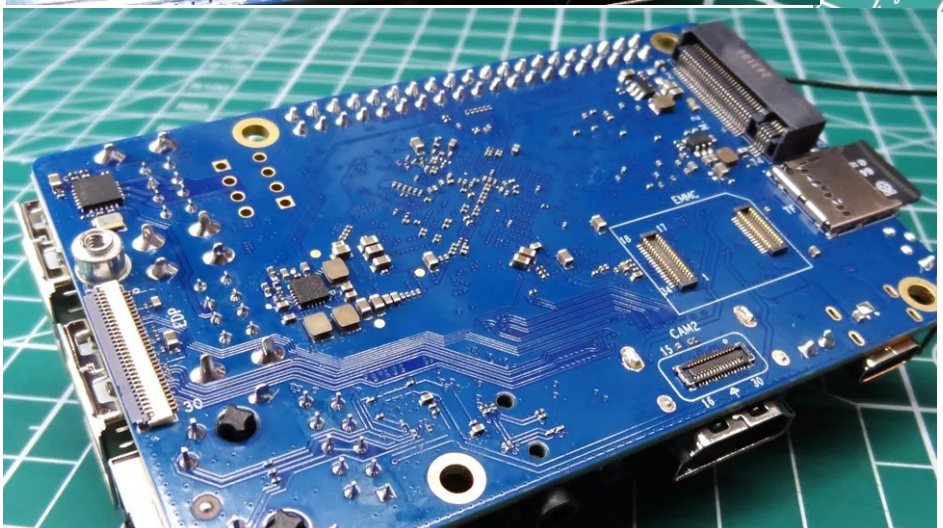
There's a dedicated 5V fan connection, so it's possible to add cooling, but it's something worth remembering in the event of long-term, heavy use.## Is it a Raspberry Pi killer? Honestly no. The Orange Pi 4A is an interesting Raspberry Pi clone, but it doesn't simply beat the Raspberry Pi at everything. It has some genuinely attractive features, such as its eight-core processor and NVMe support. The problem is with the software and overall user experience. Raspberry Pi has spent years building a huge ecosystem around its boards, and it's difficult for competitors to recreate. Tom's Hardware also found the GPIO experience less polished than what would be seen with Raspberry Pi, and some accessories and HATs potentially requiring additional work to get running.

That doesn't mean that the Orange Pi 4A is a bad product. It's an interesting board to experiment with if you're already comfortable with Linux and single-board computers, but Raspberry Pi is still probably the better choice for someone new to SBCs.## My verdictThe Orange Pi 4A is a Raspberry Pi clone with a few tricks of its own. Eight CPU cores, NVMe support, a 2TOPS NPU and plenty of connectivity makes it look very impressive on a specification sheet. Using the board shows that specifications aren't everything. The software ecosystem, compatibility and little quirks are just as important as the processor. I'd recommend it to experienced Raspberry Pi users who want to experiment something different, but newcomers might be better off remaining with Raspberry Pi.★★★★★

Rating: ★★★☆☆ — 3.5/5A powerful and interesting Raspberry Pi clone, but one with some rough edges.











What's happening At the moment!

Ziptino iPhone Four S
15d • Genius Bar (Tech Support)

iPhone 6 tethered downgrade to iOS 9

Does anyone know any working options for this???? If so please tell me (I HAVE A 6 NOT A 6S!!!)

3 4 Last comment 9d ago

Alex Smith
10d • Classic Gear

The iPhone 4 makes a great music player!

The DAC apple put in this is amazing. It makes my 40 dollar bluetooth headphones sound better than my 200 dollar headphones. Its surprising for a 16 year old phone Also side one, the iPhone 4 is old

2 0

Keaton Rozema
Aug 28 • New(er) Tech

I got a Flipper Zero



What's some things you all have done with one if you have one?

2 5 Last comment 12d ago



Linux 7.4 To Support The Raspberry Pi 10-Inch Touch Display 2

Raspberry Pi has been making it easier to build your own little computers, tablets and smart displays, and now Linux is getting an important update for one of its newest pieces of hardware. Linux 7.4 is set to add mainline support for the Raspberry Pi 10-inch Touch Display 2, meaning the display will be properly supported by the standard Linux kernel rather than relying on Raspberry Pi-specific additions. The support was added to the DRM-Misc development code ahead of the Linux 7.4 merge window. So, what's actually new? Raspberry Pi launched the 10-inch version during the summer of 2026. What makes this development interesting is what it's doing on the Linux side. The display uses a MIPI-DSI connection and an Ilitek IL79600A controller. Linux 7.4 is getting the necessary Device Tree support to recognise and properly handle this hardware. In normal-person terms, this means Linux is learning how to properly talk to the new screen. And that's actually quite important for the Raspberry Pi community. A lot of people use Raspberry Pis for projects that aren't traditional desktop PCs - things like dashboards, kiosks, control panels and homemade tablet-style computers. The 10-inch screen is particularly useful for these projects because it gives you much more space to work with than the smaller versions.

A surprisingly capable little screen

The 10-inch Touch Display 2 has a 1200 × 1920 resolution, supports 10-point multitouch, and can be used without a traditional mouse thanks to touchscreen controls and an on-screen keyboard. It's also powered directly by the Raspberry Pi rather than needing its own separate power supply. The screen is designed to be used in portrait orientation by default, although it can also be rotated into landscape mode. Raspberry Pi's documentation even provides controls for changing the orientation and brightness. Why should anyone care about a kernel update? At first, a Linux kernel update doesn't sound particularly exciting. It's not the sort of thing that gives you a completely new desktop or suddenly makes your Raspberry Pi twice as fast. But hardware support is one of those things that you don't really notice until it's not there. When a new device works properly with the mainline Linux kernel, developers have an easier starting point for building different Linux distributions and projects around it. It also means the hardware isn't quite as dependent on custom patches or Raspberry Pi-specific software. For people who enjoy experimenting with Linux and Raspberry Pi hardware, that's a pretty big deal.

A new screen for new projects

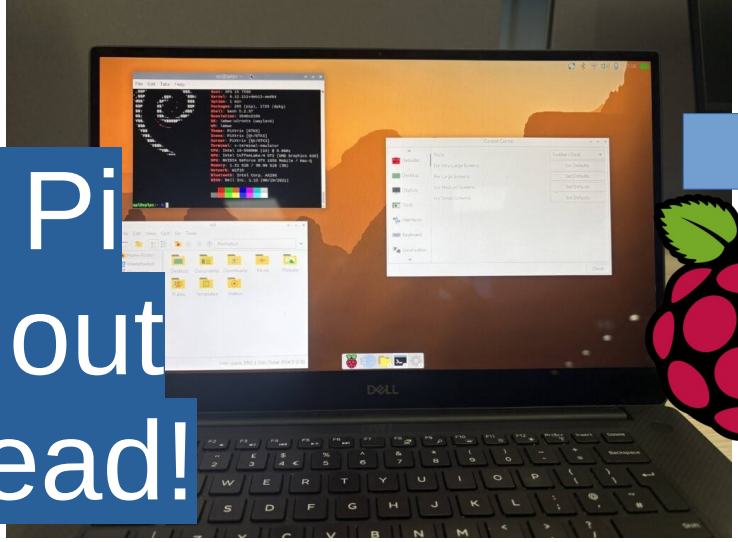
The 10-inch Touch Display 2 is clearly aimed at people who want to build something rather than simply plug a Raspberry Pi into a normal monitor. Raspberry Pi itself highlights uses such as tablets, entertainment systems and information dashboards. Personally, that's what makes this update interesting. The display isn't just a bigger monitor - it's another building block for Raspberry Pi projects. Linux 7.4 may not be the most exciting-looking update, but its new display support is a good example of something happening underneath the surface that can make a big difference to makers and Linux developers. And as Raspberry Pi continues releasing new hardware, having that hardware supported directly by the Linux kernel will become increasingly important.

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https://www.phoronix.com/news/Linux-74-RPi-10-Touch-Display-2

https://www.raspberrypi.com/documentation/accessories/touch-display-2.html

Raspberry Pi desktop is out from the dead!



After a near-four-year hiatus, Raspberry Pi Desktop for PC and Mac is back — bringing the Raspberry Pi desktop OS to ordinary computers. The new version brings the Raspberry Pi desktop environment up to date, but why did it take so long to appear? Let's dive into the details of the latest release.

An overdue return

The last release of Raspberry Pi Desktop for PC and Mac was in 2022, and it was based on Debian 11 Bullseye, being featured in the MagPi Issue 53 in a DVD. Since then, it seems that no further versions of this OS for PC have appeared, leaving the community to wonder if development had finished. However, the new release uses Debian 13 "Trixie". So, it appears that the folks at Raspberry Pi have been busy during this period updating and improving the OS.



Figure 1: The MagPi bundle

Why did it take so long?

One reason why development on Raspberry Pi Desktop for PC and Mac appears to have stalled is that Raspberry Pi makes its money by selling Raspberry Pis, not by providing OSes for x86 and Apple computers. As such, the company has little interest in maintaining a product that does not contribute directly to its bottom line — especially when the engineers working on it could be focusing on other, more important, tasks. That said, there appears to have been enough demand for a new release — as evidenced by the Raspberry Pi Foundation's tweets asking users to direct any questions about the PC version to their GitHub page. This suggests that the company received a steady stream of messages asking when a new version would be released. Perhaps the sight of all these messages motivated the foundation to begin work on this project anew.

New features

The new build contains many of the recent updates that have appeared in the officially supported Raspberry Pi desktop OS. This includes the new dock and control centre introduced in the latest Raspberry Pi OS release this year. Meanwhile, a feature called Raspberry Pi Connect allows remote access to a computer running Raspberry Pi Desktop, in addition to Raspberry Pi computers accessed in the same way. One major change is that this release is built upon a 64-bit version of Debian, meaning that 32-bit PCs are no longer supported — a trend that reflects the wider move away from 32-bit computing in recent years. According to Raspberry Pi, most computers purchased in the past decade or so should be able to run the new release. The only major exception is Apple Silicon Macs, which are not supported — due to the experimental nature of Debian's port to Apple's processors. As such, Raspberry Pi Desktop 64-bit is available for Intel-based Macs, as well as other x86-based PCs.

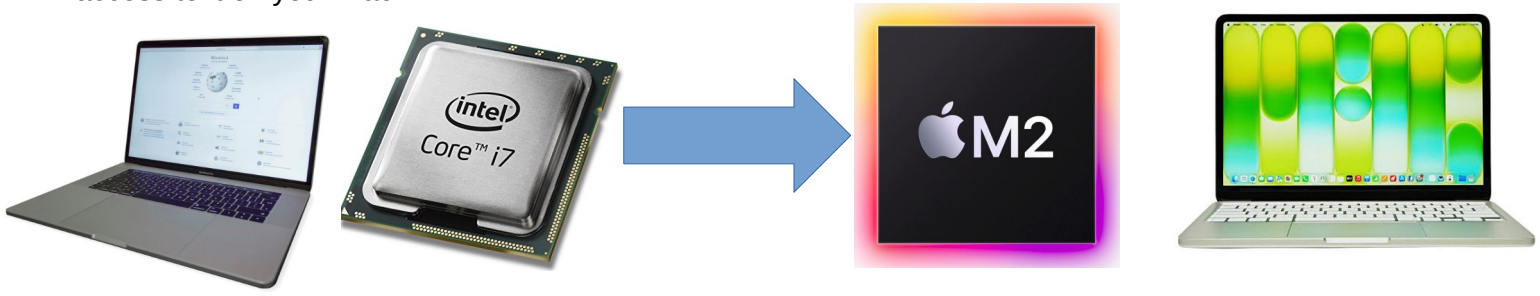
A live USB build

One benefit of the Raspberry Pi Desktop initiative is that, despite its name, you do not actually need a Raspberry Pi to use it. Instead, users can opt to try out the OS via a live USB build, which allows one to experiment with the system before choosing to install it on a more permanent location. This suggests that the release could also be a good fit for users with old computers who want to try out a lightweight Linux desktop environment. The Raspberry Pi desktop OS aims to provide all the same features and software as one might expect from a standard x86 Linux distribution — while also providing a simple, easy-to-use interface that is familiar to those used to the Raspberry Pi OS. After nearly four years since its last release, Raspberry Pi Desktop is back and better than ever. Whether or not it will continue to be updated on a regular basis remains to be seen — but, for now, it appears that this particular branch of the Raspberry Pi ecosystem is far from dead.

Why should you upgrade to a Silicon Mac

By **Lift Excelsior**

I'm sure we're all well aware by now of the time Apple announced the single most seismic piece of news in the entire history of the company since the two Steves signed off the original Apple I all those years ago. Much fanfare and brainy business chat occurred, the bottom line being, bye bye Intel, hello Apple Silicon. The M1 MacBook Air and Pro promised an otherworldly leg up in performance over the outgoing 2019 Intel models, a noticeable price reduction, incredible battery life and far less heat, and you bet it delivered! Five generations named after five BMW models later, the Apple Silicon MacBook remains the de facto choice for anyone looking for a great quality powerful productivity laptop. But why, you ask?First, let's talk about the processor. If you have a regular, run of the mill Windows PC, odds are that it has an x64 based processor. The x64 CPU architecture is descended from the great grandad x86 design, which debuted in some of the very first Intel chips. For decades, it remained the go-to choice for any desktop or laptop computer. The x64 standard did a fine enough job; the processors had decent power and can and could handle a multitude of different types of workload. However, there then arrived a competitor that would change the entire game. Enter ARM.The acronym stands for Advanced RISC Machine, RISC being an alternative instruction set to x64/x86 which uses fewer discrete instructions to perform all the exact same tasks. Having previously been used exclusively in mobile phones and tablets, in 2020, a little known someone by the name of Tim Cook had the brilliant idea to place one inside a laptop, and the results were dumbfounding.Today's M chips, together with their Snapdragon and other Windows ARM counterparts, can embarrass even the most elite of legacy x64 processors. In my MacBook Pro, I have the base 10-core M4 ARM processor, which may sound diminutive when compared to expensive gaming PCs, but in terms of performance, the revolutionary new instruction set means it can absolutely keep tabs in productivity tasks with even the most serious high end Windows computers out there. Never once in my entire 1.5 years owning the computer have I encountered a situation in which the CPU felt stressed or underpowered, something that happened rather too frequently on the 2015 i5 MacBook Pro it replaced, and an impressive statistic for someone who regularly does a lot of demanding video and audio production and other intensive tasks.On first glance of the specs of the new Macs on Apple's website, it might not seem worth it on the surface. The hefty Apple-taxed price tag only gives you 16 GB of RAM and 512 GB of SSD space as standard, as I have in my laptop. Any Windows PC for similar money would destroy this on raw specs. However, a Mac can do much more with the same set of numbers, and even in 2026, 16 GB is not a small amount of RAM by any means.You've likely heard it joked about all around the world that Windows 11 has become an incredibly bloated, inefficient operating system, bursting at the seams with AI slop, predatory data collection and increasing lack of support for still perfectly competent legacy systems. As a lifelong Apple fanboy, how much of this is true is outside of my knowledge base, but the level of polish, streamlining and everyday performance afforded by the Mac is simply impossible to beat.The latest macOS is so well refined and runs immaculately on the Apple Silicon hardware. Apps and folders open instantly, with seamless, lag-free animations. Updates can be scheduled to install overnight, rather than being forced on you right in the middle of something. For those of us with other Apple products, such as myself, you can sync data from your iPhone, iPad or Apple Watch and enjoy immediate access to it on your Mac.



Why you should upgrade to a silicon Mac

Reviews

Even the simple things, such as when you open the laptop to wake it from hibernation, the lock screen appears as soon as you lift the lid, skipping the few seconds of delay you would experience on the equivalent Windows machine or x64 Mac. Not to mention the looks of the OS are Aston Martin levels of beautiful, and despite the operating system using up to 8 GB of RAM at idle, somehow the system manages to stay blisteringly fast at all times, alluding to the simply ludicrous performance offered by the M chip. For a Windows machine to match this, you would likely need double the specs, potentially more. When I first unboxed mine in May of 2025, it honestly took me quite a while to adjust to the speed, after having spent so long on a slow and ageing Intel Mac.But my favourite thing about the latest MBP? Surprisingly, it's not the space age performance, nor the refinement of the OS. I'm sure if you dug deep enough and spent enough money, you could probably find a Windows machine that could rival it in terms of raw speed. However, there is still one discipline in which Apple still stands completely untested. Ergonomics.It's a user's very first impression of their new laptop. Touching the case, lifting the lid, typing on the keyboard, clicking with the trackpad. So many rival computers, even those worth several thousands of euros, simply cannot compare. New MacBooks feel so solidly built, the keyboard clunky and satisfying, the trackpad the best in class. In fact, the trackpad is so good, I'm not sure I could daily drive a single other one. For example, a few weeks ago I was leant a HP ProBook for a short term job. By no means a weak laptop, it did the job perfectly fine, but the trackpad was nigh-on intolerable to use. Mac trackpads support multi touch gestures, mechanically click down confidently and reassuringly, and you can click virtually anywhere without having to press a separate button or click in only a certain area of the 'pad. Not to mention the sheer size of the trackpad, meaning you'll take a while to run out of space.Media playback as well. My M4 has a gorgeous 14-inch OLED screen. A few months ago, I had to HDMI into my TV to play a YouTube video as the smart TV box was acting up at the time, and I found myself looking at my MacBook's screen far more than the TV, because of just how crisp and high resolution it is. The anti reflective coating makes it perfect for use outside, both at night and in the day, something I do frequently. And then we have the speakers. The speakers on a modern MacBook have to be heard to be believed. You hear every part of a song with ease, whether it's a quiet country ballad or a screaming UK drum and bass anthem. No Bluetooth speaker or headset I've ever experienced has beaten my laptop's speakers.Does this all sound too good to be true? Kinda. For many buyers, the base Irish price of close to €2000 would certainly be a talking point, but for such a complete package, if you have funds to spare, it would absolutely be worth it. Not to mention Macs remain hopeless at serious computer gaming, but the only video game I tend to play is Minecraft, and that runs perfectly fine on an iPad, so as an exclusively productive computer user, personally I am completely unaffected by that.These are just a few reasons why I believe you should upgrade to an Apple Silicon MacBook. Whether it's a cheap second hand M1 Air or a brand new, fully loaded M5 Pro, the results will simply blow your mind.

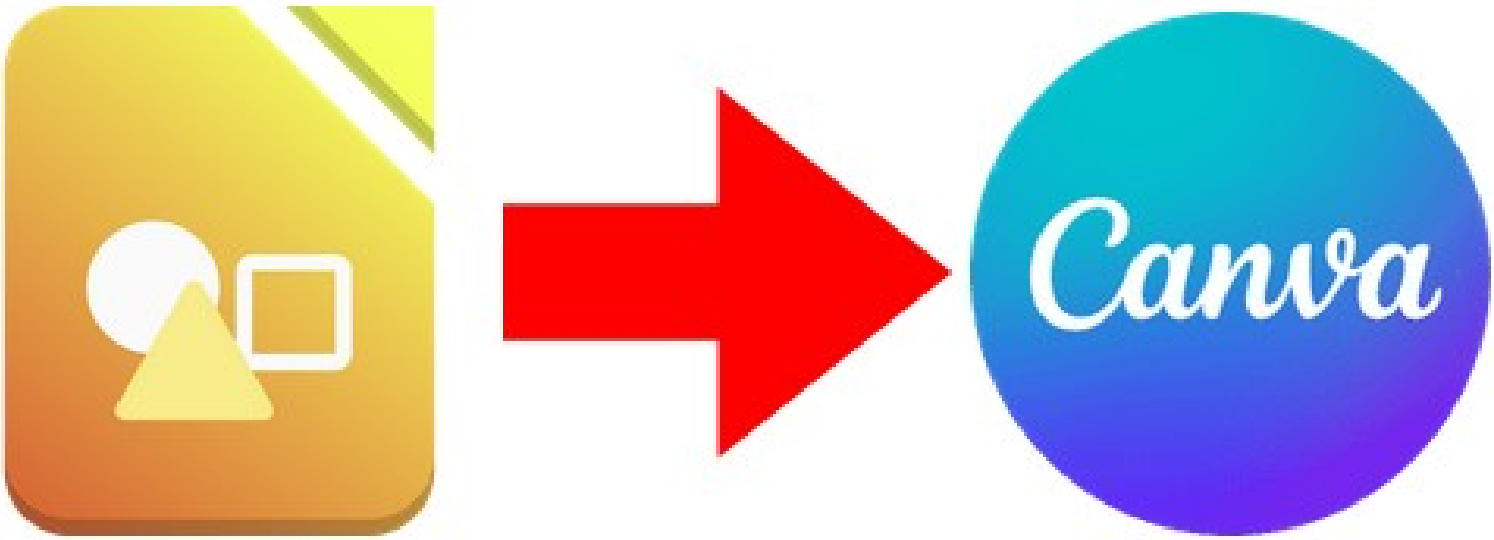


Lift Excelsior

“Hoarding imperfect PCs since lord knows when…”

A passionate tech enthusiast who lives and breathes computers, retro gear, gaming, and all kinds of hands-on experimentation

Massive Redesign!



For the past issue , we've been using LibreOffice to make our magazine. This is extremely slow to save and make edits on because it was laggy. We are switching now to Canva. Canva is very convenient. It is available on the web, as well as an app. It autosaves quickly and there is no lag. There are minor visual differences due to software.



Windows K2: Microsoft’s Plan to Fix Windows 11

For the last few years, Windows 11 has seen a wealth of new features added to it - and not all of them have found universal acceptance. Performance issues, inconsistent bits of the interface, reliability and Microsoft's own growing focus on AI have all attracted criticism. Now, Microsoft seems to be trying something new. Internally, Microsoft is working on an initiative they call Windows K2, a long-term programme aimed at improving the foundations of Windows 11, rather than simply adding more features.

What exactly is Windows K2? Despite the name, Windows K2 isn't a new version of Windows; it's an internal initiative that covers multiple Windows updates and focuses on improving the operating system itself. According to reporting from Windows Central, Microsoft has begun putting together the project during the second half of 2025; the aim is to address some of the biggest complaints about Windows 11 and make the operating system faster, more reliable and more polished. Microsoft's work is reportedly centred around three areas: performance, reliability and craft.

There's also a fourth focus: community. Microsoft wants to rebuild relationships with Windows enthusiasts and Insiders by paying more attention to feedback and providing more ways for the community to communicate directly with the Windows team. Making Windows feel faster. One of K2's biggest goals is performance. While Windows 11 has often felt surprisingly sluggish on modern hardware, things like opening File Explorer, using context menus or interacting with parts of the desktop can include delays that should be largely imperceptible on a modern PC. Microsoft is now working on reducing some of this unnecessary overhead; Windows Central notes that the company has specifically identified performance problems in areas including File Explorer, games and parts of the Windows interface. Microsoft has begun delivering some of these improvements; for example, the May 2026 Windows 11 preview update included performance and responsiveness improvements described as part of the K2 effort.

A more consistent Windows interface. Performance isn't the only thing being targeted. Windows 11 still has parts of the operating system from different generations of Windows. Some newer areas use Microsoft's modern WinUI framework, while other parts use older technology. K2 aims to reduce this inconsistency; Microsoft is working on moving more parts of Windows to WinUI 3, while also improving the framework itself. The goal is to make things like Start, the Taskbar and other system interfaces to feel more responsive and more consistent. Microsoft is also bringing back some of the customization options that users have asked for; for example, the ability to move the Windows 11 Taskbar to different sides of the screen.

Reliability over rushing features. Perhaps the biggest change behind K2 isn't something you're likely to see on the desktop at all. Microsoft is reportedly changing how its Windows teams work; the company wants to prioritize the quality of features rather than their speed of release. New features are expected to reach higher quality standards before being made part of public preview builds. The idea is that Windows shouldn't have to sacrifice reliability just to get new features out faster. This could be particularly important as Microsoft continues to add AI features to Windows. Instead of always releasing new features and leaving old parts of the operating system unfinished, K2 is intended to put the fundamentals first.

What about AI? Interestingly, K2 isn't simply an anti-AI project. Microsoft is still heavily investing in AI-powered Windows features; K2 reportedly involves reducing unnecessary AI integrations and ensuring that AI doesn't get in the way of the fundamental Windows experience. The aim is to make Windows feel like a solid operating system first, with additional features (including AI) built on top. This is important because of the recent Windows controversy. Microsoft is also working with developers to encourage more native Windows applications built around modern frameworks such as WinUI 3. The company believes that encouraging more native apps on Windows could help improve performance and consistency across the board.

Windows K2: Microsoft’s Plan to Fix Windows 11

Is Windows 12 coming? K2 has also raised questions about Microsoft's future plans for Windows. However, K2 is itself far from being Windows 12, and there's no need to think of it as being a secret new operating system at the moment. It's better to see it as Microsoft trying to improve Windows 11 from within; that could make it far more significant than another numbered Windows release. Instead of abandoning the current platform and starting over again, Microsoft appears to be working to fix the inconsistencies that have built up over time.

Why K2 matters

Windows is used by hundreds of millions of people; even seemingly small problems can become incredibly frustrating when they repeat themselves every day. A slow context menu might only waste a fraction of a second. A sluggish File Explorer might only delay you for a moment. An inconsistent interface might only be mildly annoying. But when they're repeated hundreds of times, they become noticeable. That's why Windows K2 could be one of Microsoft's most important Windows projects in years. It's not about a flashy new logo or a redesigned desktop. It's about making Windows work better. If Microsoft can deliver on its promises, the result could be a Windows 11 that feels faster, more reliable and more consistent - while still looking familiar. Perhaps the biggest Windows update of the next few years could be the one that makes Windows itself better.



This article is sourced from trusted websites so that our information is trusted and safe. These websites are: videocardz.com.

What Windows version should you use?

While Windows 11 looks much the same, Microsoft does in fact offer several different editions of its operating system. Some are geared towards everyday use, while others provide features for business, education and powerful workstation PC's. I explain the differences, and whether it's worth paying more for a better edition of Windows.

Windows 11 Home Edition

This is the edition you're most likely to find pre-installed on a new laptop or desktop PC. It provides the essential features required by the majority of users, such as the Start menu, Microsoft Store, gaming support, Windows Security and everyday productivity tools. For the purpose of web browsing, gaming, video playback and schoolwork, Home is generally completely sufficient. However, there are some limitations compared to Windows Pro that may be of interest to power users, such as the inability to act as a Remote Desktop host or manage BitLocker encryption. It is good for gaming and casual use with bloadware and ads and forced updates.

Windows 11 Pro Edition

Windows 11 Pro builds upon the features of Home and provides more advanced system management options that should appeal to power users. This includes BitLocker Drive Encryption, the ability to host Remote Desktop sessions, Hyper-V virtualisation support and the option to join domains or Microsoft Entra ID. Hyper-V in particular is a useful tool that allows developers and tech enthusiasts to run virtual machines on Windows, enabling them to test other operating systems without the need for another physical computer. For most home users however, Pro will not magically make games run faster or make their PC more powerful - the main benefits are focused on additional features and not improved day-to-day performance. It gives more control of the PC. Still a bit of bloatware but not much.

Pro for Workstations Edition

Windows 11 Pro for Workstations is aimed at users requiring more advanced technologies for heavy lifting PC's. This includes ReFS file system support and persistent memory on compatible hardware. This edition is not recommended for the everyday user and should only be pursued by those wishing to use specialised applications and high-end hardware. It isn’t that good

Enterprise Edition

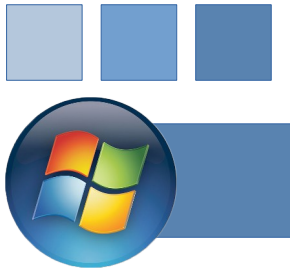
Windows 11 Enterprise is a further step above Pro and is primarily aimed at businesses managing large-scale IT deployments. This edition includes additional features and technologies enabling improved security, deployment and administration of Windows clients across an enterprise network. These features are generally beneficial to large-scale organisations. Enterprise is a volume licensing edition of Windows designed to be deployed by an organisation's IT department, and is generally not available to individual consumers. It has no bloatware. If your company has some spares then it is really worth it or if you want to buy in bulk.

Education and Pro Education Editions

These editions are essentially tailored versions of Windows 11 for educational institutions. Similar to Enterprise, they provide tools that enable administrators to securely configure, deploy and manage Windows devices within a school environment. Most of these features are not available to general consumers, however students may find themselves using a PC running Education as their primary operating system, without the need to purchase a Pro edition themselves. It is only available for schools. If your school is willing to hand out a spare.



What Windows version should you use?



Windows 11 LTSC Editions

Windows 11 LTSC stands for Long-Term Servicing Channel and provides a version of Windows tailored to meet the needs of industrial and commercial PC's. Unlike other editions, LTSC does not recieve regular feature updates and is only provided every few years. This is primarily aimed at manufacturers and developers of industrial PC's, to allow a stable base operating system that does not require regular servicing. This includes medical devices, digital signage, public transport systems and self-checkout machines in stores. The standard Windows 11 Enterprise LTSC 2024 edition recieves support until October 2029, with Windows 11 IoT Enterprise LTSC 2024 recieving support until October 2034. IoT Enterprise is the industrial variant of LTSC and is intended for fixed-purpose PC's. LTSC should not be viewed as a special edition of Windows for everyday PC's. While it does recieve security and quality updates, it has a much longer servicing period and does not follow the same pattern of new feature updates as the standard editions of Windows. Additionally, LTSC has different licensing options and lacks some of the traditional consumer features you would expect from an edition such as Home or Pro. Windows 11 LTSC is a bloatware free Windows edition and your first choice if you don't urgently need new AI features that were released recently and don't mind waiting a bit for updates.

Windows 11 SE Edition

While not officially an edition of Windows 11 in the traditional sense, Windows 11 SE was a simplified version of the operating system aimed at education devices and low-cost PC's. These were intended to be deployed in a managed environment, rather than for general daily use. However, Windows 11 version 24H2 is the final supported release for SE and should not be considered as an alternative to Home or Pro. It can not open .exe files…. Don't use it unless if it is the ONLY thing that can save your PC.

Which Windows 11 Edition Should You Choose?

As you can see, most people should be perfectly happy with Windows 11 Home. Pro is a worthwhile upgrade for those that specifically require features such as Hyper-V, BitLocker management, Remote Desktop hosting or domain management. Enterprise and Education are primarily for organisations, while Pro for Workstations should only be used by those with specific high-end PC needs. The key thing to remember is that a higher edition of Windows will not magically turn your PC into a powerhouse - you'll still need a capable processor, sufficient memory, decent graphics hardware and fast storage for everyday tasks and gaming. The different editions simply open up a range of advanced features and management options.

So here's my decision..

Home: For casual people and gamers that just want to open an app/game and play and check emails. Check if you're ok with forced updates. Bitlocker and most remote desktop apps won't work

Pro: Unlike home, you have more control of your PC. There are still ads. Remote desktops and bitlocker works.

Enterprise: Gives an ad free experience and removed bloatware and forced updates. However, you can only buy in bulk. If your company has leftovers or don't mind buying bulk, this is your goto option for non bloatware.

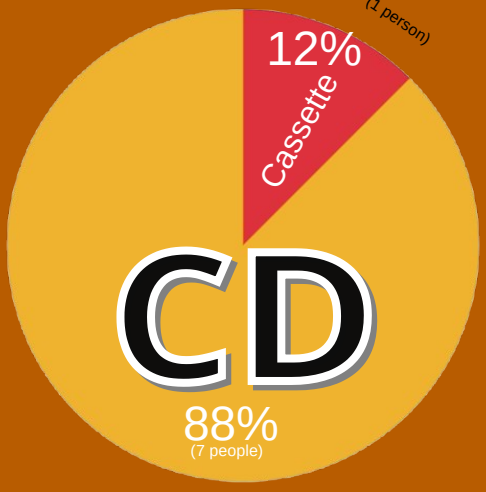
LTSC: Same as Workstation with no bloadware or ads, however, updates are delayed. This is your goto except if you need really recent features (e.g. Copilot AI features). Goto if you can't get Workstation or education.

Education: Don't be fooled by the name. Same as workstations but for schools. There is also no ads in Microsoft Edge because they want students to not feel distracted. This isn't available to consumers and only to schools but if you see one lying around, it is worth it.

SE: Do not buy! It is only made for really slow laptops to compete with Chromebooks. It cannot run EXE files, apps run in fullscreen with no minimise and take up less than GB of RAM.

The Tech Debate

Last issue...



Cassette or CD walkman?

android or iOS





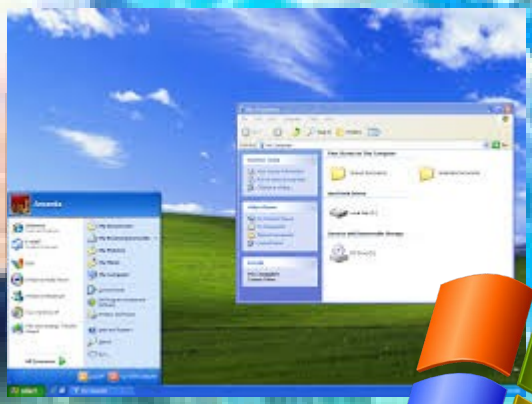
Cast your vote in by the Legendary Tech skool community or:
https://tinyurl.com/techmagazinedebate
No sign-in required

The history of Windows Frutiger Areo

If you used a computer during the 2000s, chances are you've seen Frutiger Aero used, even if you don't know what it is. Bright blue skies, green grass, glossy buttons, transparent windows, bubbles, fish and glossy icons. Frutiger Aero was the style of technology trying to make the future seem exciting, and friendly. Frutiger Aero style is now known as Frutiger Aero, a name that was created years after the style became popular. The name combines Frutiger, which refers to the Frutiger typeface family, and Aero, Microsoft's visual design language that was introduced with Windows Vista. The aesthetic is generally associated with around 2005 to 2013.

One of the important steps towards this style was Windows XP, which was released in 2001. XP introduced the colourful Luna interface, with its blue title bars, green Start button and colourful icons. Microsoft itself described XP as having a noticeably revised user interface, including a new Start menu and taskbar. However, Microsoft didn't develop its visual style completely on its own. Apple had introduced Aqua with Mac OS X, using glossy buttons, transparency, shadows and highly detailed icons. Microsoft was clearly aware of Apple's design, and an internal Windows XP theme called “Candy” was discovered years later that looked remarkably similar to Aqua. It was never released as the final XP theme, but it shows just how much Apple's interface influenced Microsoft's designers. The final Luna design was different, but you can still see some of the same ideas: bright colours, rounded shapes, gradients and a more playful appearance.





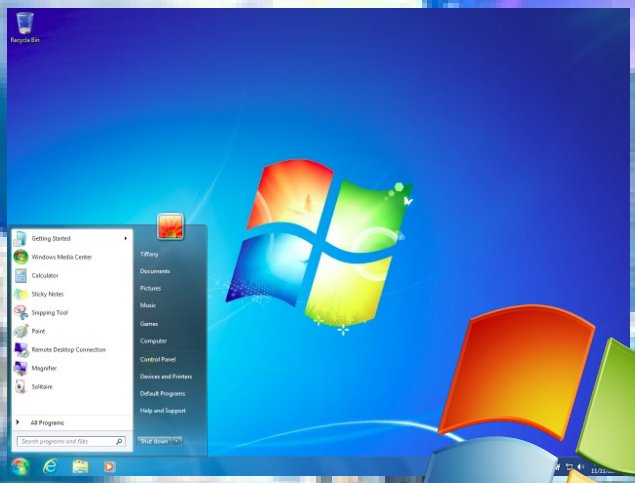
In 2007, Microsoft released Windows Vista. The new Windows Aero style became much closer to what we now call Frutiger Aero. Vista's desktop had translucent glass effects, reflections and smooth animations. Its Desktop Window Manager could render windows separately and combine them to create the famous frosted-glass effect. Vista also introduced features such as Flip 3D.The design wasn't just about making Windows look shiny. Microsoft wanted the interface to feel more modern and open. Microsoft described Aero's glass windows as creating an "open, energizing environment."



The history of Windows Frutiger Areo

Frutiger Aero wasn't just Windows Aero. It was a much bigger design trend across technology, advertising, websites and consumer products. Designers started to combine glossy technology with images of nature. Blue skies, water, grass, plants, bubbles and tropical fish became common. The idea was almost like technology and nature living together. This helped create the strange but memorable look of the late 2000s: a computer could have a transparent glass interface while its wallpaper showed a perfect green field and blue sky.

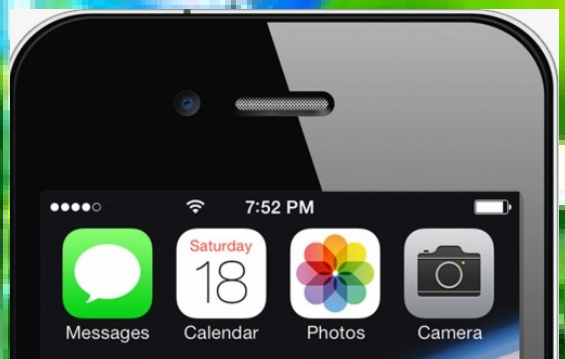
Windows 7 arrived in 2009 and continued the Aero design, although in a more refined way than Vista. Transparent window borders, colourful icons, glossy effects and the redesigned taskbar all fit into the same visual world. By this point, similar design ideas could be found everywhere. Websites, phones, game consoles, advertisements and even product packaging often used gradients, reflections, bubbles and bright colours.



By the early 2010s, technology started to move in a different direction. Flat design became increasingly popular. Instead of realistic textures, shadows and glossy buttons, designers began using simpler shapes, fewer colours and flatter icons. Microsoft eventually moved away from Aero with Windows 8, while Apple and other technology companies also reduced the amount of skeuomorphic detail in their interfaces. Today, Frutiger Aero has become a nostalgic aesthetic. The Consumer Aesthetics Research Institute describes it as a style associated with glossy design, glassy materials, nature imagery and humanist typography.

What makes Frutiger Aero interesting is that it represented a very specific idea of the future. Technology was shown as something exciting, colourful and connected to nature. Modern interfaces are usually much simpler and flatter. Looking back at Windows XP, Vista and Windows 7 can therefore feel like looking at a version of the future that never happened. For many people, that is exactly what makes Frutiger Aero so memorable.





They’re Targeting Your Home Computer Now

Are we losing control of the computers we own? For decades, a personal computer meant owning a machine that was truly yours. You could install your own software, store your documents on it and customise it as you see fit. However, as computers become more connected to online services and artificial intelligence, some tech-savvy individuals are beginning to wonder if we will ever truly lose ownership of our personal device. Now, technology commentator Loyal Moses discusses how the computer industry wants to push everyone towards a future where big companies control everything you can do on your personal devices. He then goes on to cite several changes in Windows, cloud services and modern computer architecture to substantiate his claim.

Is Windows becoming less private?
One of the biggest headaches for Windows users is the amount of data that Microsoft collects about them. Windows 11 collects diagnostic data to help secure the OS, resolve issues and improve future versions of Windows. Optional diagnostic data can also be used to collect additional information about how you browse online. Fortunately for users, Microsoft provides a list of all these collected data that can be toggled off in Windows. Microsoft's Recall feature has also been a topic of conversation in terms of privacy. Recall lets you search through a record containing snapshopped content of the screen, and it is intended for use on compatible Copilot+ PCs, but is considered an opt-in feature. This means that only those who actively turned on snapshot saving will have their screen contents analysed and stored by the PC. Microsoft also assures that this content is processed locally and not uploaded into the cloud for analysis. Even with all these controls and options for users to turn off specific features, features that record or analyse activity bring with them many questions. Does what the OS records pose a threat to my privacy? Is the information about these features' activities easily accessible and clear? Are privacy-friendly settings easy to find?

The cloud versus your own computer.
Another issue brought up is the increasing amount of computer operations that require cloud services. Various artificial intelligence applications and services use large-scale servers to process information, which means doing more online than on your personal PC. While using remote servers could certainly be useful in certain situations, having everything rely on a single entity's infrastructure, conditions and availability could be very limiting. Another alternative is to host files and processes on a local device depending on the type of software used. One specific example of this is running artificial intelligence models locally. A capable PC can handle some AI models without having to rely on an external service to process every query, which can offer much more control over one's data, albeit with higher system requirements than an online model would need.

Does your hardware really belong to you?
Finally, we touch on various aspects of hardware security, such as Secure Boot, Trusted Platform Modules (TPMs) and Microsoft Pluton. These are all technologies designed to improve the overall security of a computer by preventing unauthorised access or tampering, but critics of these technologies believe that these security features can also be used to restrict what operating system or software can run on a specific PC. The reality is more nuanced than an either/or situation, with Secure Boot being more of a security feature that prevents malicious programs from running at startup while TPMs are used to secure features like encryption and random number generation that are vital to the security of Windows. Another factor to consider when discussing hardware security is repairability, especially with the trend of newer computers to have soldered-in memory and storage, which can make repairs and upgrades harder to do. For people who want to keep using a computer for as long as possible, having easily replaceable hardware can be a huge boon.

What can users do?
The artcles's central message is to demand responsibility for your own digital sovereignty. Subscribing to privacy-conscious features, careful selection of software, keeping documents backed up and exploring if running a local version of a particular service will suffice are all good ideas to explore. However, disconnecting from the internet or buying a used computer may not be the best ideas, since older operating systems are much more likely to have vulnerabilities that modern malware can target, and having a computer not connected to the Internet can also expose it to attacks via infected files or physical media.The issue at hand is one of control. A computer should be a tool that enhances the life of its owner while also being a system that the user knows and understands. It should not become a device that is controlled almost exclusively by one company through its online services and dictate what one can and cannot do. While the industry's intentions are still up for debate, privacy, repairability and user autonomy should always be priorities.

Halloween Projects

Welcome to the dark side of inventing! In this special feature, we will be guiding you on to how to make the best Halloween projects! From a Lego Mindstorms candy shooter, Raspberry Pi light up pumpkin to an arduino falling table! This guide has it all! So read on to see how to make the most grew some and meanest of inventions!

LEGO CANDY SHOOTER



Credits: JK Brickworks

This LEGO candy shooter shoots out candy wherever! It is powered by a LEGO Mindstorms EV3 brick. However, you can replace it with an Arduino or another MINDSTORMS product such as NXT or RCX. It uses the EV3 Infrared sensor to detect when something is in front of the machine to trigger the delivery mechanism. The chocolates are stored in a gravity feed stack and are pushed out from the bottom using a simple piston arm. Once they are pushed out, they land on a platform that is rotated up to throw them. The feed system was designed to be adjustable, so it can accommodate a range of these mini chocolate bars. Or pretty much anything that will fit and stack reasonably well.

The official website of the person who made it is here. You can then download the instructions and watch a demo video about it: jkbrickworks.com/halloween-chocolate-thrower/

Light-up Raspberry Pi decoration



Credits: ModMyPi

It's time to grab the most terrifying Raspberry Pi board ever devised this Halloween!

The assembled Halloween PumpkinPi is a programmable LED board designed for any maker or hacker. It's a fun option for those new to coding with the Raspberry Pi, or even experienced users who want a bit of fun this scary season!

The Halloween PumpkinPi is fully assembled (no soldering required) but you'll need to code in your LED pattern to get your pumpkin terrifying the neighbours!

Falling Table!



Credits: Mark Rober

Halloween candy thieves beware! Mark Rober has created an awesome trick using a dropping table.

This trap was made by Mark Rober for his Candy Thieves vs Rigged Candy Bowl Halloween video. The collapsing table will drop when someone tries to steal all the candy in the bowl. There are many other engineering pranks like this one.

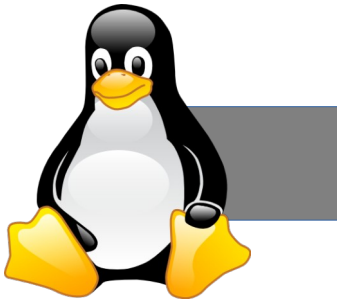
Watch Mark Rober's video here: https://www.youtube.com/watch?v=Zb01RStdzEs

PLEASE NOTE: We did not make any machines or contribute in any part of them. All credits go to the owner or company labelled in each project



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Linux alternatives to Mac and Windows apps



Now is the perfect time to start switching over to Linux. There have been Linux distributions that look identical to windows for almost 20 years now, so now is the perfect time to do it! But the problem with all of this is that there are barely any applications that have support for Windows, Mac AND Linux, which is terrible, so today, this article will guide you with the best and correct alternatives for these applications so you don't have to spend hours researching!

Microsoft Office --> LibreOffice

If you are looking for a free alternative to Microsoft Office, then LibreOffice is one of the best options available on Linux. It includes Writer for documents, Calc for spreadsheets and Impress for presentations, meaning you can do almost everything you would normally do with Microsoft Office. The interface is slightly different, but it is still fairly easy to learn. LibreOffice can also open and save many Microsoft Office file formats, so you can continue working with documents made on Windows without needing to buy Microsoft Office.





LibreOffice®

MS Paint/Paint.net --> Photopea

If you need only to edit photos, there are several alternatives to Paint on Linux. However, you will not find a program as easy to use as Paint. The site Photopea .org is a good alternative because it works directly in the browser without the need to download and install the program. This utility has many features such as layers, crop, resize, filters, text, and much more. Photopea also allows you to open Photoshop files to make more complex edits. However, if you only want to make simple drawings, it is better to use a simpler program.





Photoshop --> Virtual Machine

These are all of the alternatives we could find. Unfortunately, even though there are Linux alternatives to Adobe Photoshop, the apps do not have the advanced editing features of Photoshop. The only alternative we could find is Photopea but it is not advanced enough to be worthy. However, if you do just want to paint basic stuff, then you should go to photopea or if you are not switching to Linux, just use paint.net. The only and the best way to get Photoshop on Linux is to run a virtual machine of Windows 10/11 and download it that way.





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Coming (hopefully) 1st December